



Pharm*Achieve*

SEIZURES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE	Adverse Effect	NHP	Natural Health Product
AEDs	Anti Epileptic drugs	OXC	Oxcarbazepine
BMD	Bone Mineral Density	PB	Phenobarbital
BRI	Brivaracetam	PER	Perampanel
CBZ	Carbamazepine	PHT	Phenytoin
CNS	Central Nervous System	PRM	Primidone
EEG	Electroencephalogram	SE	Side Effect
COC	Combined Oral Contraceptive	SJS	Stevens Johnson Syndrome
ETHO	Ethosuximide	STI	Stiripentol
GBP	Gabapentin	TEN	Toxic Epidermal Necrolysis
Hx	History	TOP	Topiramate
KD	Ketogenic Diet	VPA	Valproic Acid (or divalproex)
LAC	Lacosamide		
LEV	Levetiracetam		
LTG	Lamotrigine		

LEARNING OBJECTIVES

- 1 Identify the types of seizures based on symptoms
- 2 Understand the stepwise approaches to drug therapy
- 3 Understand the interactions between anti-epileptic drugs and hepatic enzymes
- 4 Understand the interactions between anti-epileptic drugs and contraceptive methods
- 5 Learn how to manage adverse effects of AEDs
- 6 Identify appropriate drug therapies for special populations

DEFINITIONS

Seizure	An acute episode defined by a sudden uncontrolled change in electrical signaling in the brain that can cause changes in behavior, movement or consciousness
Epilepsy	- A chronic neurological disorder characterized by a predisposition to unprovoked seizures, where a patient has: - A history of at least two unprovoked seizures occurring > 24 hours apart - A singular unprovoked seizure AND a high ($\geq 60\%$) probability of further seizures over the next 10 years - A singular unprovoked seizure AND an identifiable epilepsy syndrome demonstrated by electroencephalogram (EEG) abnormalities
Status Epilepticus	- Seizure lasting > 5 minutes - Seizure episodes occurring in rapid succession without patient recovering between them

PATHOPHYSIOLOGY

↑ Expression of proconvulsive neuropeptides + change in ratio of inhibitory and excitatory receptors in synaptic clefts + ↓ inhibitory neuropeptides



↑ excitability resulting in 1st seizure



Seizure-induced neuronal death causes changes in long-term gene expression



↓ threshold for subsequent seizures



Neuronal reorganization in favor of excitability

CAUSES, RISK FACTORS AND TRIGGERS



Causes and risk factors may overlap

Causes

- Genetics
- Infection (meningitis, encephalitis)
- Hypoxia
- Stroke (ischemic/hemorrhagic)
- Head trauma
- Electrolyte disturbances (↓ Ca, ↓ Mg)
- Hypoglycemia OR severe hyponatremia
- Idiopathic

Triggers

Hyperventilation, stress, sensory stimuli, photostimulation, sleep deprivation, hormonal changes

Risk factors

- Most common in childhood and age > 65
- Men more likely
- Family History
- Prolonged seizures during childhood
- Alcohol & drug withdrawal/toxicity
- High fever in children
- Cerebral hemorrhage
- Drugs that ↓ seizure threshold (e.g. bupropion)
- Illicit substances (e.g. cocaine)
- Renal or hepatic failure

HISTORY AND CLINICAL FEATURES

FOCAL (PARTIAL) SEIZURES

Focal seizures **without** impairment of awareness
(simple partial seizures)

- **NO impairment in awareness**
- Features: motor, sensory, autonomic, cognitive features
- **Duration: < 60 seconds**

Focal seizures **with** impairment of awareness
(complex partial seizures)

- **Impairment of awareness**
- Features: same as simple partial; pause of activities/**blank stare**
- **Duration: 1-2 minutes**

Focal to bilateral tonic-clonic
(secondarily generalized tonic-clonic)

- **Impairment of awareness**
- Start as focal seizure with impaired awareness **then progresses to bilateral convulsive** features such as tonic-clonic spasms

- Focal seizures originate in **one hemisphere of the brain**
- Is aura present? **Possibly focal (partial) seizure**
- Background analysis: Ask patients if they have a current or history of seizures, stroke, intracranial infection, head trauma, medical conditions affecting CNS or neurologic disorders, family history of seizures, drug use or withdrawal

HISTORY AND CLINICAL FEATURES

GENERALIZED SEIZURES

Absence (petit mal)	• Impaired consciousness • Features: staring • Duration: 5-10 seconds • Onset in childhood/adolescence
Generalized tonic-clonic (grand mal)	• Impaired consciousness • Features: tonic phase (rigid muscles) followed by clonic phase (rhythmic jerking muscles) with possible tongue biting, incontinence, cyanosis • Duration: 1-2 minutes
Atonic	• Possible impaired consciousness • Loss of muscle tone (drop attacks) • Duration: <15 seconds • Common in children
Myoclonic	• NO impaired consciousness • Brief bilateral "shock-like" jerks or jerking of a group of muscles • Duration: <5 seconds

*Magnetic Resonance Imaging (MRI), Computed Tomography (CT) scanning, and electroencephalography (EEG) are important diagnostic tests that may be performed in order to aid in the diagnosis of epilepsy

GOALS OF THERAPY

- ✓ Manage patients experiencing their first seizure
- ✓ Prevent or reduce the risk of seizure recurrence in individuals diagnosed with epilepsy
- ✓ Minimize or prevent the adverse side effects of antiepileptic drugs (AEDs)
- ✓ Enhance quality of life through patient education

NON-PHARMACOLOGICAL MEASURES

Lifestyle changes

- Maintain consistent sleep routines
- Avoid flashing lights in photosensitive patients
- Minimize use of proconvulsant substances e.g. alcohol, cocaine, and amphetamines

Patient education

- Educate patients on the safe use of medications and NHPs
- Discuss seizure precautions with patient and family
- Discuss driving restrictions- check your province (e.g. Ontario: 6 months seizure free)

Complementary therapy

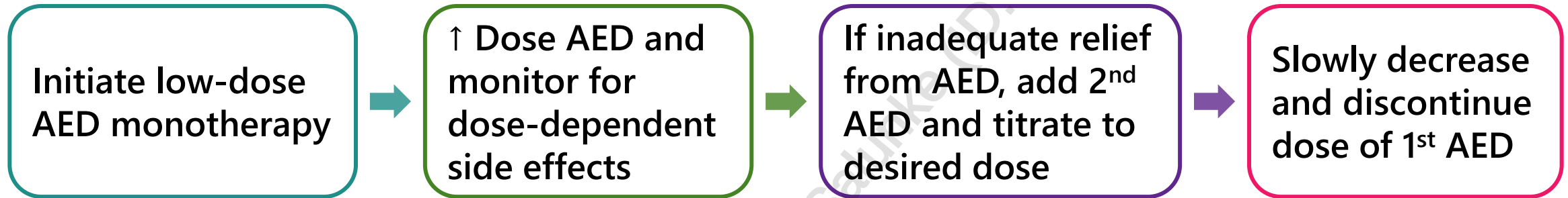
- High-fat, low-carb, ketogenic diets (evidence in drug-resistance epilepsy)
- Vagal nerve stimulation via electrical impulses or deep brain stimulation via leads
- Yoga, mindfulness-based therapy may improve quality of life and seizure control
- Meditation, acupuncture and relaxation training have limited evidence

PHARMACOLOGICAL MEASURES CONCERNS

Some concerns with evidence-based guidelines in epilepsy:

1. No standalone Canadian guidelines available
2. Variability and different interpretations of available evidence by various international guidelines
3. Many medications are used off-label before evidence is available → used in practice but not in guidelines
4. Quality and heterogeneity of evidence varies → lower confidence in recommendations

TREATMENT ALGORITHM & TIPS



The choice of AED is based on:

- Seizure type
- Patient population
- Comorbidities
- Drug interactions
- Side effect profile

Start LOW & go SLOW



AED polytherapy is reserved for patients who fail to achieve acceptable control on 2-3 monotherapy drugs

PHARMACOLOGICAL MEASURES

Seizure Types	Focal Seizures	Generalized Seizures				Other
	Focal with/ without impairment & Focal to Bilateral Tonic-Clonic	Absence*	Tonic or Atonic*	Myoclonic*	Tonic-Clonic	Status Epilepticus
First Line Therapy	C – CBZ/ OXC L – LTG L – LEV L – LAC	ETHO	V – VPA L – LTG	V – VPA L – LEV	V – VPA L – LTG L – LEV	IV Lorazepam IV Midazolam IV PHT IV Propofol
Second Line Therapy	T – TOP V – Vigabatrin P – PHT P – PB P – PER Eslicarbazepine	V – VPA L – LTG L – LEV	CLOB TOP Rufinamide (by epilepsy specialist)	CLOB TOP L – LTG BRI	CLOB TOP PER BRI	IM Midazolam Rectal Diazepam

*Avoid the following AEDs in the seizure types with an asterisk due to possible worsening: CBZ, OXC, GBP, Vigabatrin; PHT avoided in absence and myoclonic seizures

MEMORY TOOLS: FIRST-LINE THERAPY

Focal: LLLOC

- Focal → Focus → Look → **LLLOC**: lamotrigine, levetiracetam, lacosamide, oxcarbazepine, carbamazepine

Absence: Ethosuximide

- Common in children → Children **suck** on their thumb → Etho**sux**amide



Tonic-Clonic**ic**, Tonic-Atonic**ic** and Myoclonic**ic** ALL have Valproic acid as first-line options.

You just need to memorize the other first-line options, which includes lamotrigine and levetiracetam

Tonic-Clonic: Lam + Lev

- Valproic acid
- Clonic → Clone → 2 options → Lamotrigine, levetiracetam

Tonic-Atonic: Lam

- Valproic acid
- Atonic → A → The L option with "a" → Lamotrigine

Myoclonic: Lev

- Valproic acid
- The one that's left → Levetiracetam

CAM [Clonic-Atonic-Myoclonic]

C → Lam + Lev

A → Lam

M → Lev

HOW DO AEDs AFFECT HEPATIC ENZYMES?

Enzyme-inducing AED

Reduced efficacy of hepatically metabolized drugs e.g. COCs

- CBZ
- Eslicarbazepine
- Oxcarbazepine
- PER (≥ 8 mg/day)
- PB
- PHT
- PRM
- Rufinamide
- TOP (≥ 200 mg/day)
- Clobazam (weak)

Non-enzyme-inducing AED

Do not affect hepatically metabolized drugs

- Brivaracetam
- ETHO
- GBP
- Lacosamide
- LTG
- LEV
- VPA
- Vigabatrin

EFFECTS OF AEDs ON CONTRACEPTIVE METHODS

Contraceptive methods likely to be LESS effective in women taking enzyme-inducing AEDs

Combined contraceptive patch

Combined oral contraceptive

Progestin implant

Progestin-only oral contraceptive

Vaginal ring

Contraceptive methods **NOT** likely to be affected by enzyme-inducing AEDs

Barrier contraception

Depot progesterone

Intrauterine device (hormonal or nonhormonal)

MANAGEMENT OF ADVERSE EFFECTS OF DRUG THERAPY

ADVERSE EFFECT	DESCRIPTION	MANAGEMENT
Dose dependent effects	CNS and GI effects Dizziness, sedation, fatigue, ataxia, cognitive and psychiatric symptoms, nausea. Higher risk with CBZ, LTG, PRM, tiagabine, TOP, VPA	- Reduce dose of AED - If related to peak dose (30-60 minutes after each dose) of IR-release drug, consider slow-release or frequent low-dose administrations of IR AED or bedtime once daily dosing
Idiosyncratic	Skin rashes - Most likely to occur within first 6-8 weeks of therapy - May progress to SJS or TEN Higher risk with PHT, CBZ, LTG	Stop AED immediately after definite drug rash (→life-threatening reactions if continued) - Consider AED with low risk of skin rash that's not chemically related - Genetic testing for HLA-B*15:02 recommended for patients from China, Thailand, Malaysia, India
Chronic Effects	Low bone density and fractures Higher risk with enzyme-inducing AEDs and VPA	Monitor bone density and consider supplementation of vitamin D and calcium as per osteoporosis guidelines
Other reactions	Blood and/or hepatic dyscrasias Elevation in liver enzymes and/or reduction in blood counts Higher risk with carbamazepine (neutropenia) and valproic acid (thrombocytopenia)	- Obtain baseline CBC and serum liver aminotransferases, repeat in 4-6 weeks - If levels remain stable, reduce to yearly blood work

PHARMACOLOGICAL MEASURES- BARBITURATES

	Phenobarbital	Primidone
Mechanism	Potentiates GABA's effect on the GABA receptor, increasing Cl ⁻ channel activity	
Dose dependent side effects	CNS, hyperactivity	CNS, behavioural changes
Chronic side effects	Behavioural and cognitive problems, mood changes, sedation, depression, diminished libido, anemia, folate deficiency, osteomalacia	
Idiosyncratic side effects	Rash, hepatotoxicity	
Metabolic pathways	- Metabolism: CYP 2C9/2C19 - Induces: CYP 3A4/1A2/2B6/2C9 25% renal elimination	- Metabolism: CYP 2C19 - Induces: CYP 3A4/1A2/2B6/2C9 40-50% renal elimination
Comments	- Long half-life (once daily dosing) - VPA inhibits metabolism ↓efficacy of hormonal contraceptives	- Metabolized to PB (PRM also has clinical effects) but short half-life (QID dosing) - VPA inhibits metabolism ↓efficacy of hormonal contraceptives

PHARMACOLOGICAL MEASURES- BENZODIAZEPINES

	Clobazam
Mechanism	Potentiates GABA's effect on the GABA receptor, increasing Cl ⁻ channel activity
Dose dependent side effects	CNS
Chronic side effects	Behavioural, cognitive, and mood changes
Idiosyncratic side effects	N/A
Metabolic pathways	• Few drug interactions • Metabolized by CYP 3A4, 2C19, 2B6 • Weakly induces CYP 3A4 • 2 metabolites: norclobazam is an active metabolite with 1/4th potency compared with parent versus 4-hydroxyclobazam which is the inactive metabolite • Metabolites eliminated renally (94%)
Comments	• Tolerance develops (initial good response followed by loss of seizure control) • Rapid onset, can be used as add-on for patients "nearly seizure free" • Withdrawal symptoms if not tapered off

PHARMACOLOGICAL MEASURES- CARBOXYLIC ACID DERIVATIVES

	Valproic acid/divalproex
Mechanism	Inc. GABA activity, Na ⁺ and K ⁺ channels
Dose dependent side effects	CNS, GI upset, tremor, thrombocytopenia, sedation
Chronic side effects	PCOS, menstrual cycle abnormalities, weight gain, hyperammonemia, reduced bone mass, less commonly hyponatremia
Idiosyncratic side effects	Hepatic toxicity (more likely in first 6 months), pancreatitis, alopecia, blood dyscrasias (thrombocytopenia)
Metabolic pathways	• Metabolism: CYP 2A6/2B6/2C19/2E1 • Inhibits: CYP2C9 • Renal elimination: 2%
Comments	• Teratogenic – avoid in women of childbearing potential • DDI: Carbapenems ↓ VPA levels & ↓ seizure threshold → uncontrolled seizure risk

FYI

What is the difference between valproic acid and divalproex?

- Divalproex (DVP) is converted to valproic acid (VPA) in the gastrointestinal tract. It is an *enteric-coated* (coated for stomach protection) ester preparation of VPA. It causes less stomach upset than VPA
- VPA is more cost-effective than DVP

PHARMACOLOGICAL MEASURES

GAMMA AMINOBUTYRIC ACID (GABA) DERIVATIVES

	Gabapentin	Vigabatrin
Mechanism	Binds to voltage-gated Ca^{2+} channels	Inhibits GABA-T (including GABA levels)
Dose dependent side effects	CNS, tremors, vision changes	CNS, permanent vision loss
Chronic side effects	Weight gain, tremors, vision changes	Permanent vision loss
Idiosyncratic side effects	Pedal edema	Anemia, peripheral neuropathy, weight gain, hyperactivity, mood changes (e.g. depression)
Metabolic pathways	No CYP metabolism Renal elimination: 90%	No CYP metabolism Renal elimination (unchanged): 80%
Comments	• Add-on drug for Focal Seizures • Can use in liver failure (not metabolized hepatically) • No significant drug interactions • Dose dependent bioavailability	• Few interactions, BID dosing, no dermatologic, hepatic, or hematologic effects • Permanent visual field defects limits use • May worsen absence or myoclonic seizures

PHARMACOLOGICAL MEASURES- HYDANTOIN DERIVATIVES

	Phenytoin
Mechanism	Inhibits voltage-gated Na ⁺ channels
Dose dependent side effects	CNS, dose-related encephalopathy
Chronic side effects	Behavioural problems, skin thickening, folate deficiency, gingival hyperplasia, hirsutism, coarsening of facial features (acne), cognitive impairment, metabolic bone disease, nystagmus, increased liver enzymes
Idiosyncratic side effects	Blood dyscrasias, rash (rarely severe)
Metabolic pathways	• Metabolism: CYP 2C9/2C19 • Induces: CYP 3A4 (strong) + 1A2/2B6 • Renal elimination: 5%
Comments	• Dosing is complicated by saturation kinetics • Therapeutic serum concentration range: 10-20 mcg/mL (40-80 µmol/L) • Dose conversion required when switching between formulations • Daily or BID dosing

PHARMACOLOGICAL MEASURES- IMINOSTILBENE DERIVATIVES

	Carbamazepine	Oxcarbazepine	Eslicarbazepine
Mechanism	Blocks voltage-gated Na ⁺ channels		
Dose dependent side effects	CNS (e.g. ataxia, nystagmus)	CNS	CNS
Chronic side effects	Hyponatremia, neutropenia, gastric side effects	Hyponatremia (> CBZ), neutropenia (< CBZ)	Hyponatremia, PR interval prolongation
Idiosyncratic side effects	Rash, blood dyscrasias, hepatotoxicity	Rash	Rash
Metabolic pathways	- Metabolism: CYP3A4 - Induces: CYP3A4 (strong) + 1A2/2B6/2C9 < 1% renal elimination	- Metabolism: CYP3A4 - Induces: CYP3A4 (weak) - Inhibits: CYP2C19	- No CYP metabolism - Induces: CYP3A4 (mod) - Inhibits: CYP2C19 (weak)
Comments	- IR or CR dosing - May worsen absence seizures, exacerbate/produce myoclonus - Mild anticholinergic effects	- Higher risk of hyponatremia (vs CBZ) - Lower risk of neutropenia (vs. CBZ) - Minimal enzyme induction	- Once daily dosing - Lowest risk of hyponatremia - Minimal enzyme induction - Only indicated for Focal (Partial) Seizures

PHARMACOLOGICAL MEASURES- SUCCINIMIDE DERIVATIVES

	Ethosuximide
Mechanism	Inhibits T-type calcium channels
Dose dependent side effects	CNS, GI upset
Chronic side effects	Behavioural problems
Idiosyncratic side effects	Rash, blood dyscrasias
Metabolic pathways	• Metabolism: CYP3A4 • Renal elimination: 12-20% • Few drug interactions
Comments	• Only indicated for Absence Seizures

PHARMACOLOGICAL MEASURES- SV2A

	Levetiracetam	Brivaracetam
Mechanism	Binds to vesicle protein SV2A (Brivaracetam has higher affinity)	
Dose dependent side effects	CNS, behavioural problems (LEV > BRI)	
Chronic side effects	-	
Idiosyncratic side effects	Psychosis, serious dermatological reactions	
Metabolic pathways	- No CYP metabolism - Renal elimination: 66%	- Metabolism: CYP2C19 - Renal elimination: 95%
Comments	- Rapid titration - May be used in pregnancy for generalized tonic-clonic seizures but levels may decrease	Needs more support evidence

PHARMACOLOGICAL MEASURES- OTHERS

	Lacosamide	Lamotrigine
Mechanism	Inhibits voltage gated Na ⁺ channels	Binds to Na ⁺ , Ca ²⁺ , HCN channels
Dose dependent side effects	CNS, increased PR interval prolongation on ECG	CNS, rash, possibly insomnia
Chronic side effects	Behavioural problems	-
Idiosyncratic side effects	Increased LFTs	Rash (SJS)
Metabolic pathways	- Metabolism: CYP 2C19/2C9/3A4 - Renal elimination: 70%	- No CYP metabolism - Renal elimination: 10%
Comments	Caution in patients with cardiac conduction abnormalities	- Very slow titration is required - Addition of hormonal contraceptives (COC) reduces LTG serum levels by up to 50%: consider doubling LTG dose after starting COC - May be used in pregnancy for generalized tonic-clonic seizures - Similar efficacy to carbamazepine and better tolerated

PHARMACOLOGICAL MEASURES- OTHERS

	Perampanel	Rufinamide
Mechanism	AMPA glutamate receptors antagonist	Inhibits voltage-gated Na ⁺ channels
Dose dependent side effects	CNS, behavioural changes	CNS, GI
Chronic side effects	Weight gain	
Idiosyncratic side effects	Psychiatric and personality changes	Multi-organ hypersensitivity, status epilepticus, leukopenia, ↓QTc
Metabolic pathways	- Metabolism: CYP 3A4/1A2/2B6 - Possible CYP induction at high doses - Renal elimination: 22%	- No CYP metabolism - Induces: CYP3A4 (weak) - Renal elimination: 2%
Comments	- Serious psychiatric effects in patients with/without history of psychiatric conditions - Monitor during titration and at high doses - Avoid consuming alcohol - Decreases efficacy of levonorgestrel-containing hormonal contraceptives	- Indicated as adjunct in generalized seizures in Lennox-Gastaut syndrome - Can cause SE in some patients

PHARMACOLOGICAL MEASURES- OTHERS

	Topiramate
Mechanism	Binds to Na ⁺ , Ca ²⁺ channels, GABA-A, AMPA, inhibits carbonic anhydrase
Dose dependent side effects	CNS, word finding, hyperthermia and cognition problems
Chronic side effects	Weight loss, kidney stones
Idiosyncratic side effects	Metabolic acidosis, acute angle glaucoma, oligohydrosis, paresthesias
Metabolic pathways	• Induces: CYP 3A4 • Inhibits: CYP 2C19 • Renal elimination: 70%
Comments	• Cognitive effects limits use • Requires slow titration

COMPARING 1ST LINE MONOTHERAPIES

AED	Side Effect Profile	Place in Therapy
CBZ	Hepatotoxicity, anticholinergic, neutropenia, hyponatremia	Partial seizures but better options available
LTG	- SJS, decreased concentration by COC - Better tolerated than CBZ and similar efficacy (Large RCT)	Partial seizures, atonic and tonic-clonic
PHT	Gingival hyperplasia, folate deficiency, ↑ body hair	Avoid if other options are available due to many AEs and complicated dosing
LEV	- Avoid if psychiatric disorders present - Safe option in pregnancy for generalized tonic-clonic seizures	Partial seizures, myoclonic and tonic-clonic
ETHO	Milder (mostly GI) common AEs than other AED	Absence seizures
VPA	Avoid in liver and pancreatic disease, pregnancy, and women of child-bearing age	Atonic, myoclonic and tonic-clonic seizures

SEIZURES MANAGEMENT TIPS DURING PREGNANCY

- LTG levels may decrease variably during the 2nd and 3rd trimesters of pregnancy
 - Recommended to refer the patient to a healthcare provider for monthly monitoring and registration with the North American Antiepileptic Drug Pregnancy Registry
 - Therapeutic drug monitoring is necessary to account for serum fluctuations and ensure patients remain seizure-free throughout pregnancy
 - **Note:** LTG levels rise quickly after delivery (within 1-2 weeks); thus, doses must be adjusted early after delivery to avoid toxicity in the patient
- TOP, OXC, LAC, LEV levels may also ↓ during pregnancy → Monitor
 - TOP may ↑ risk of cleft palate
- **Enzyme-inducing AEDs may increase fetal vitamin K degradation;** thus, vitamin K is routinely administered to newborns after delivery to prevent hemorrhagic diseases
- **Patients should take at least 1 mg of folic acid from pre-conception until the end of the first trimester.** Then, they can switch to 0.4 mg/day to prevent neural tube defects



The drug of choice should ultimately be the drug that best controls seizures!



During pregnancy, avoid:
VPA, PB, PRM



SEIZURES MANAGEMENT TIPS DURING BREASTFEEDING

- During breastfeeding, CBZ, PHT, VPA, LTG are likely safe
 - Breastfeeding should **NOT** be discouraged;
 - Clinical significance of AED transferred into breast milk is unclear
 - AED monitoring infants is not routinely performed but can be on case-by-case bases
 - Watch for side effects in infants like sedation (benzo, barbiturates, ETHO, LTG)



The drug of choice should ultimately be the drug that best controls seizures!



During breastfeeding, use PB, PRM, ETHO with **caution**



MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Frequency of seizures	Reduce or eliminate	Days-weeks
Physical harm from seizures (falls, injuries)	Prevent	Ongoing

Knowledge of the serum level at which a patient achieved an appropriate clinical response can provide a useful reference range at future points in time such as when pharmacokinetic changes in the body occur. Maintaining the serum level within therapeutic range is not a priority; it is important to treat the patient, not the serum level.

MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Drowsiness/sedation	Eliminate or reduce to manageable/tolerable level	Duration of therapy
Weight gain		
SJS/TEN	None	Up to 8 weeks post-initiation
CBC	Within normal range	Obtain at baseline then after 4-6 weeks. Reduce to annually once stable
LFTs		



Complete a BMD after 2-5 years of AED use and before starting AED in postmenopausal women. Treat with calcium/vitamin D

WHEN TO STOP AEDS?

AEDs are typically used lifelong but factors that would favour successful discontinuation are:

- Seizure free for ≥ 2 years on stable dose of AED
- Normal neurological exam
- Normal EEG

Factors associated with a poor prognosis are:

- History of high frequency of seizures
- Multiple episodes of status epilepticus
- Combination of seizure types
- Development of abnormal mental functioning

TAKEAWAY

- Select a single AED based on seizure type → AED polytherapy if failure to achieve acceptable control on 2-3 monotherapy drugs.
- It is important to treat the patient, not the serum level. In pregnancy and breastfeeding, the drug of choice is the one that best controls seizures

First-line treatments:

1. **Focal seizures:** originate from one side of the brain
 - Lacosamide, lamotrigine, levetiracetam, oxcarbazepine, carbamazepine
2. **Tonic-clonic seizures:** characterized by muscle stiffening phase followed by rhythmic jerking
 - Valproic acid, lamotrigine, levetiracetam
3. **Atonic seizures:** characterized by sudden loss of muscle tone
 - Valproic acid, lamotrigine
4. **Myoclonic seizures:** characterized by brief, bilateral jerking motions
 - Valproic acid, levetiracetam
5. **Absence seizures:** characterized by short episodes that present as a blank stare
 - Ethosuximide

CASE 1

SE is a 34-year-old female that has been discharged from hospital after experiencing a seizure. She asks for your advice about seizure medications. The notes from the hospital state that SE had a 90-second seizure that involved bilateral jerking movements and an episode of incontinence. She has no recollection of this. After the seizure, she was groggy and tired. This was her first reported seizure. Her BP is 128/77 mmHg, her HR is 74 bpm, her BMI is 30.3 kg/m², and her EEG normal. She has no allergies to any medications. She has a medical history of hypertension and obesity for which she takes amlodipine 10 mg PO daily, levonorgestrel/ethinyl estradiol 100/20 mcg PO daily, semaglutide 1 mg SC weekly, multivitamin 1 tab PO daily, and polyethylene glycol 17 g PO daily. She does not smoke, she drinks 3-4 alcoholic drinks per day, and she works as a delivery driver.

1. What type of seizure did she have?
 - a) Absence
 - b) Generalized tonic-clonic
 - c) Complex partial seizure
 - d) Myoclonic



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2. Which medication would be the MOST appropriate to start her on?
- a) Valproic acid
 - b) Phenytoin
 - c) Carbamazepine
 - d) Levetiracetam



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3. SH reveals that she intends to become pregnant soon. Which medication would be the LEAST appropriate to start her on?
- a) Valproic acid
 - b) Phenytoin
 - c) Carbamazepine
 - d) Levetiracetam



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4. SH was not prescribed an AED on discharge as it was likely a provoked seizure as per the medical team. Which of the following is NOT a possible contributor to her provoked seizures?
- a) Her blood glucose level immediately post seizure was 2.4 mmol/L
 - b) She had 8 drinks of alcohol the day prior to presentation
 - c) Admission blood work revealed a serum creatinine of 349 µmol/L due to severe dehydration
 - d) Her blood pressure was 128/72 mmHg on presentation



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CHANGE LOG

July 2025

- Formatting changes made throughout
- Slide 5: Changed '>' to '≥' 60%
- Slide 7: Added 'genetics' to causes
- Slide 14: Edited note on the drugs that can worsen certain seizure types
- Slide 18: Added note that skin rashes can develop into SJS or TEN

September 2025

- Slide 33-34: Elaborated on monitoring parameters

Oct 2025

- Slide 14: Edited second line therapies

November 2025

- Slide 7: Addition of illicit substances and provided example of drug that can lower seizure threshold
- Slide 11: Provided example of driving restriction in Ontario and addition of variation amongst provinces
- Slide 12: Paraphrased text to improve clarity
- Slide 18: Edited onset of skin rashes to 6-8 weeks
- Slide 31: Added note on consequences of topiramate use in pregnancy
- Slide 34: Updated frequency of BMD
- Slide 36: Updated takeaway slide
- Slide 42: Updated references

April 2026

- Slide 5: Added EEG note to epilepsy
- Slide 21: Added that carbapenems decrease seizure threshold
- Slide 36: Simplified and updated takeaway